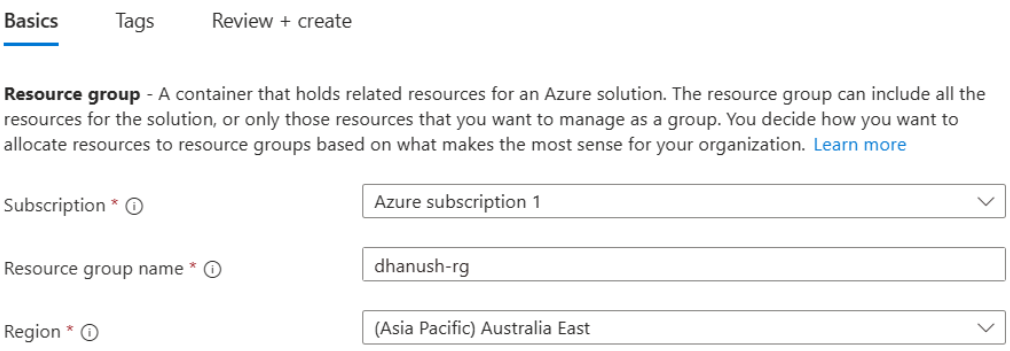
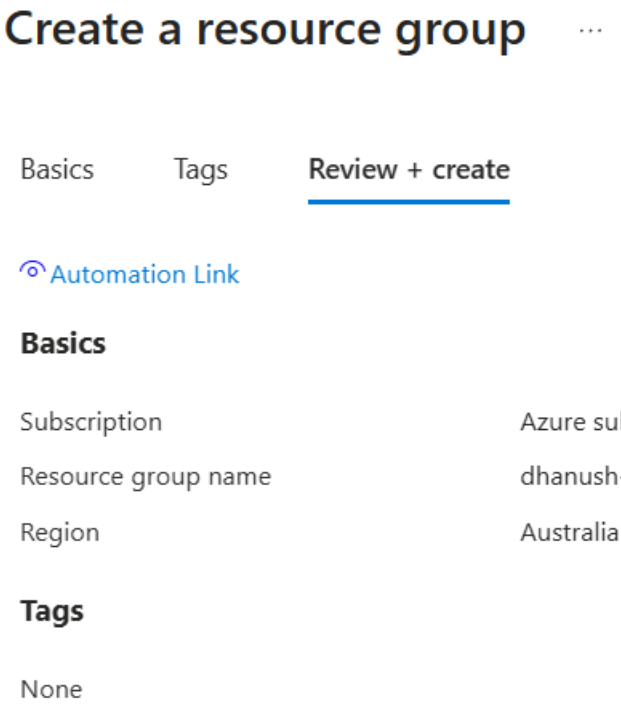
LAB 4: Create 2 (or) 3 Linux VMs in an Availability Set

STEP1 – Create Resource Group

1. Sign in to Azure Portal.
2. Go to Resource groups → Create.
    - Name: <studentname>-rg
    - Region: (e.g., *East US, Central India*)



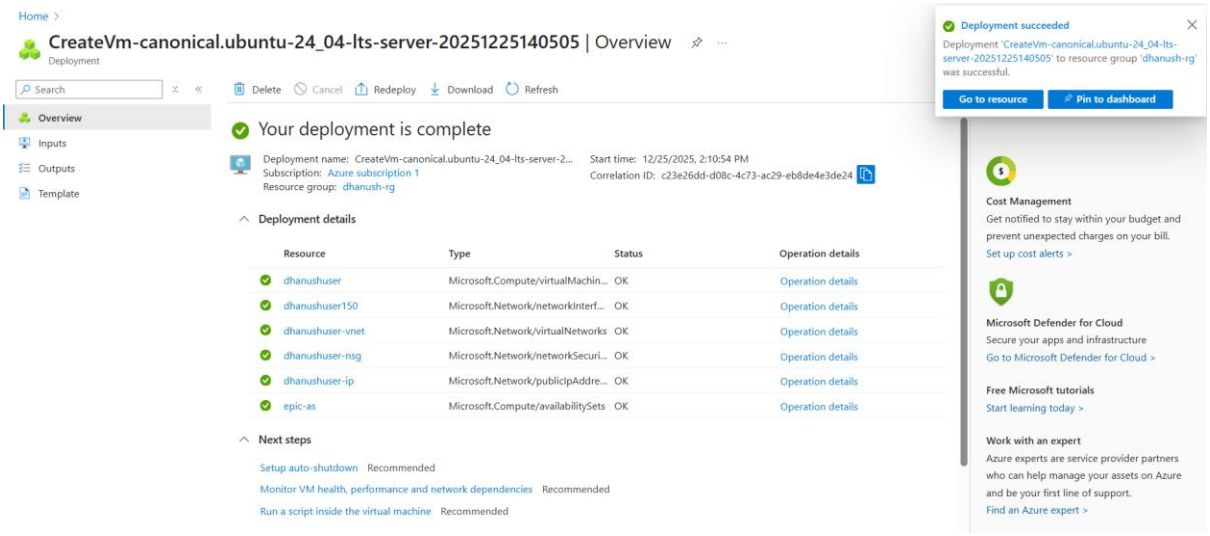
- **Review + Create → Create**



STEP2 – Create Availability Set in te vm

1. In the search bar, type Availability sets → Create
   - Resource group: <studentname>-rg
   - Name: <studentname>-availset
   - Region: same as RG
   - Fault domains: 2 (simulates racks/power units)
   - Update domains: 5 (for rolling updates

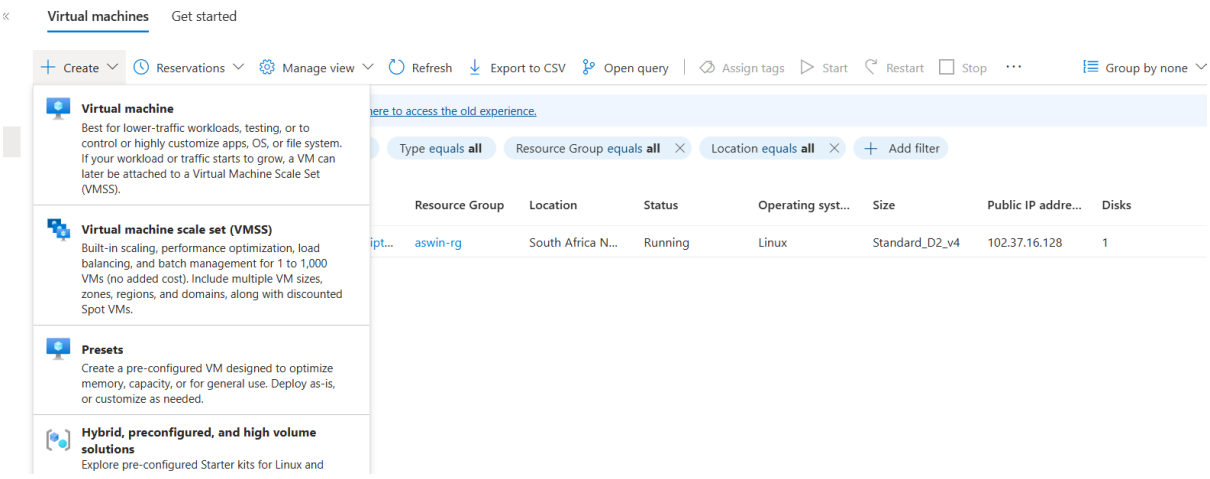
- Click Review + Create → Create



Create 3 Linux VMs inside the Availability Set

VM1

- Go to Virtual machines → Create



- Resource group: <studentname>-rg
- VM name: <studentname>-linux-vm1
- Region: same as RG

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ — Azure subscription 1

Resource group * ⓘ — dhanush-rg

Create new

Instance details

Virtual machine name * ⓘ — dhanushuser

Region * ⓘ — (Europe) France Central

Deploy to an Azure Extended Zone

Availability options ⓘ — Availability set

- Availability options: Availability set → Select <studentname>-availset
- Image: AlmaLinux OS 9 – x64 Gen2

Availability options ⓘ — Availability set

Based on your input, you might want to consider creating this resource as a virtual machine scale set, which allows you to manage, configure and scale load balanced virtual machines. Create as VMSS

Availability set * ⓘ — (new) epic-as

Create new

Security type ⓘ — Trusted launch virtual machines

Configure security features

Image * ⓘ — Ubuntu Server 24.04 LTS - x64 Gen2 (free services eligible)

See all images | Configure VM generation

VM architecture ⓘ — Arm64 / x64

Run with Azure Spot discount ⓘ

- Size: Standard_B1s
- Authentication: SSH public key
- Inbound ports: Allow SSH (22)

Inbound port rules

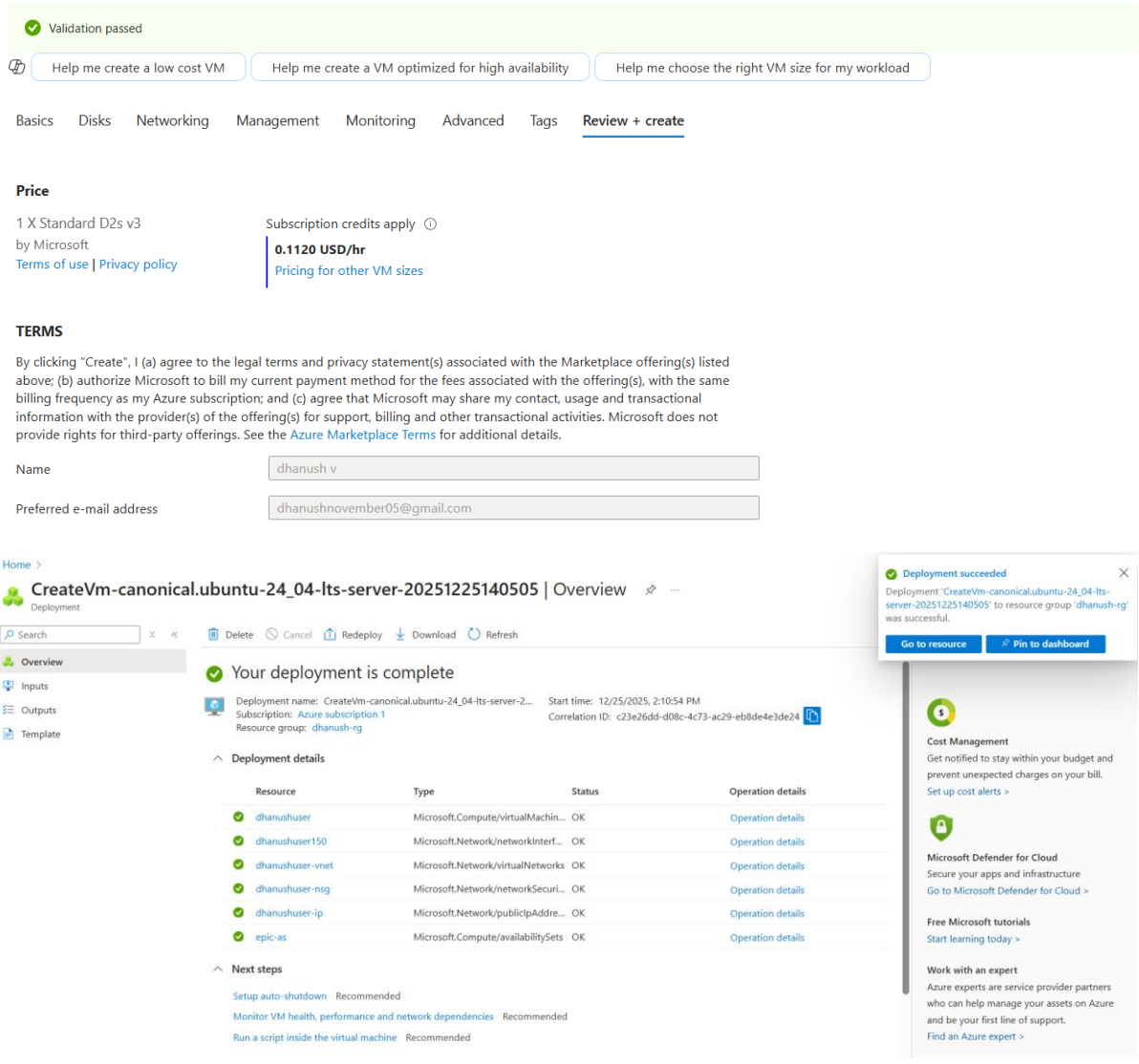
Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ
- ○ None
- ◉ Allow selected ports

Select inbound ports * : HTTP (80), SSH (22)

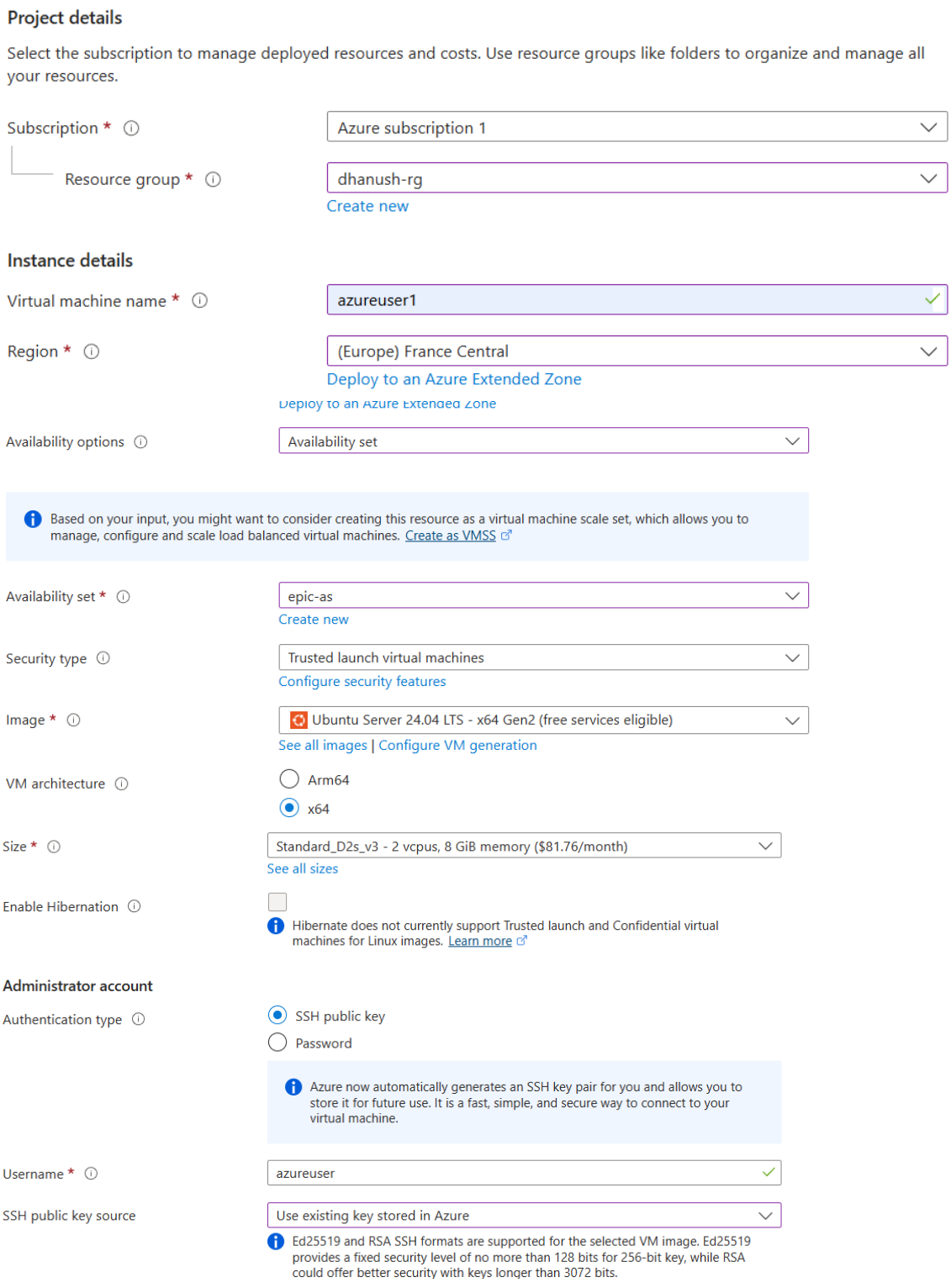
⚠ **This will allow all IP addresses to access your virtual machine.** This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

- **Review + Create → Create**



VM2

- VM name: <studentname>-linux-vm2
- Availability set: <studentname>-availset
- Other settings same as VM1



Stored Keys: ezhumalai-webserver-vm_key

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ
- ○ None
- ◉ Allow selected ports

Select inbound ports *: HTTP (80), SSH (22)

⚠ **This will allow all IP addresses to access your virtual machine.** This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

✓ Validation passed

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | **Review + create**

Price

1 X Standard D2s v3
by Microsoft
Terms of use | Privacy policy

Subscription credits apply ⓘ
0.1120 USD/hr
Pricing for other VM sizes

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.

Name: dhanush v
Preferred e-mail address: dhanushnovember05@gmail.com

CreateVm-canonical.ubuntu-24_04-lts-server-20251225142102 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name: CreateVm-canonical.ubuntu-24_04-lts-server-2...
Subscription: Azure subscription 1
Resource group: dhanush-rg

Start time: 12/25/2025, 2:23:44 PM
Correlation ID: 31681d7f-2d1a-4c10-b5a0-987c1fe40eac

Deployment details

Resource	Type	Status	Operation details
azureuser1	Microsoft.Compute/virtualMachin...	OK	Operation details
azureuser1320	Microsoft.Network/networkInterf...	OK	Operation details
azureuser1nsg254	Microsoft.Network/networkSecuri...	OK	Operation details
azureuser1-ip	Microsoft.Network/publicIpAddre...	OK	Operation details

Next steps

Setup auto-shutdown Recommended
Monitor VM health, performance and network dependencies Recommended
Run a script inside the virtual machine Recommended

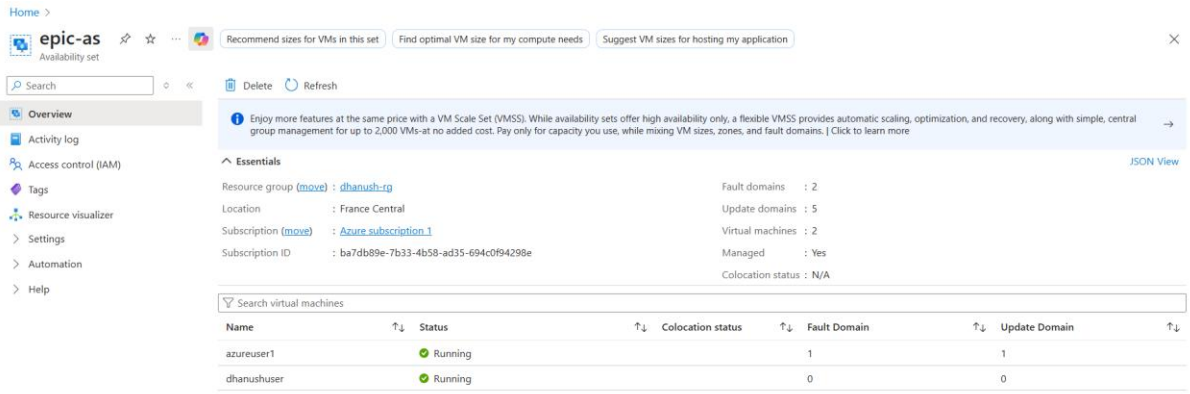
Go to resource | Create another VM

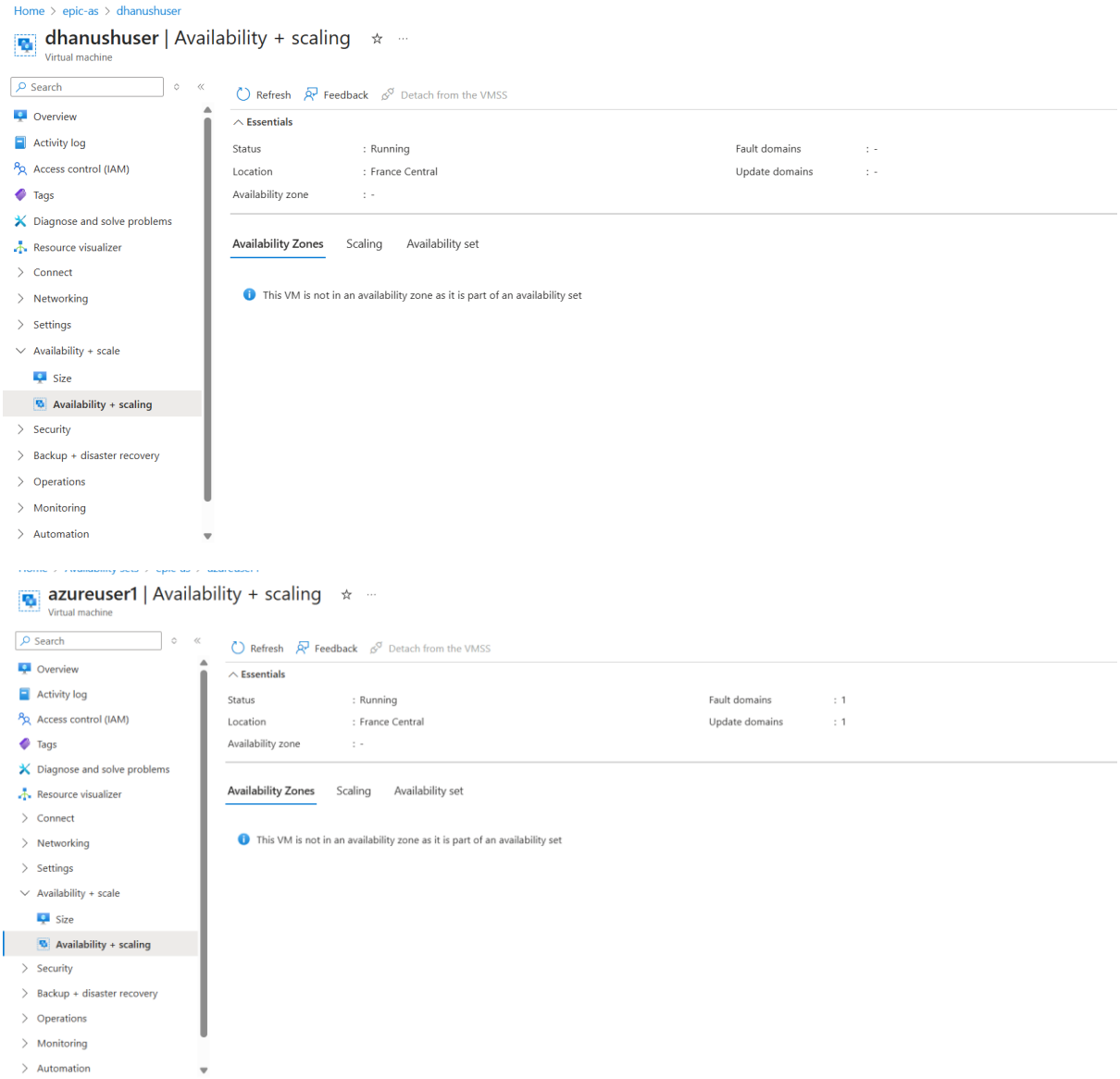
VM3

- VM name: <studentname>-linux-vm3
- Availability set: <studentname>-availset
- Other settings same as VM1

Part 4 – Verification

1. Open Availability Set → Instances
    - Confirm VMs are distributed across Fault Domains (FDs) and Update Domains (UDs).





2.Connect to each VM using SSH and check:

hostname && uptime

Outcome

- You now have 3 Linux VMs inside an Availability Set.

- Azure ensures:
- At least one VM stays up during planned maintenance (update domains).
- VMs are distributed across different fault domains (different racks/power sources).